

DSA0402 - Fundamentals
of Data Sciences

Assignment Tool - 1

... Short Answers ...



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①. What is Data science? Explain its primary objective.

Data science is an interdisciplinary field that uses statistics, mathematics, programming, and domain knowledge to extract useful information and knowledge from data.

Primary Objective:

To analyze data and discover meaningful patterns, insights and predictions that help in better decision-making.

②. List any four benefits of Data science in business and society.

Four benefits are:

- * Improved decision-making
- * Time and cost reduction
- * Increased accuracy
- * Personalized services.

③. What are the different facets of data? Briefly explain each.

The major facets of data can be explained as:

1. **Structured data:** Data organized in a fixed format such as rows and columns.

Ex: Excel spreadsheet.

2. **Semi-structured data:** Data that does not follow a strict table structure but contains tags, labels or keys.

Ex: JSON & XML

3. **unstructured Data**: Data without a predefined structured

Ex: Images, videos and text.

④. What is Big Data? state its key characteristics (5V's)

Big Data refers to extremely large, complex and rapidly generated datasets that are difficult to process using traditional data processing methods.

The 5V's of Big Data are:

1. **Volume** - Huge amount of data.
2. **Velocity** - speed at which data is generated and processed.
3. **Variety** - Different types and formats of data.
4. **Veracity** - Accuracy and reliability of data.
5. **Value** - Useful information obtained from data.

⑤. What are the major phases of the Data science process?

The major phases are:

1. Define the problem
2. Retrieve Data
3. Cleanse Data
4. Integrate Data
5. Transform Data
6. Analyze Data
7. Build Models
8. Present Findings
9. Build Application.

⑥. why is data collection considered an important step in the Data science process?

Data collection is important because the quality and quantity of collected data directly affect the analysis and model results.

- * Obtain relevant information
- * Reduce errors and bias
- * understand the problem accurately.
- * Build reliable machine-learning models.

⑦. what is data integration? why is it necessary in Data science?

Data integration is the process of combining data from multiple sources into a unified dataset.

It is necessary because it:

- * Combines information from different sources.
- * Provides a complete view of the data.
- * Removes data inconsistencies.
- * Helps in accurate analysis and modeling.

⑧. what is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining, processing and interpreting data to discover patterns, relationships and useful insights. Four techniques are:

1. Statistical Analysis
2. correlation Analysis
3. Regression Analysis
4. Clustering

⑨. what is meant by building a model in Data science ?
mention the basic steps involved.

Model building is the process of creating a mathematical or machine-learning model that learns patterns from data and uses them to make predictions or decisions.

Basic steps:

1. Collect and prepare data.
2. select relevant features.
3. Split data into training and testing sets.
4. select a suitable algorithm.
5. Train the model.
6. Test and evaluate the model.
7. Tune and improve the model.

⑩. why is presenting findings an essential part of the Data science process ? Mention any four visualization tools or techniques.

Presenting findings is important because it helps to communicate complex data and insights clearly to users and decision-makers.

Four visualization tools / techniques are:

1. Bar charts
2. Line charts
3. Pie charts
4. Scatter plots

Tools include Tableau, Power BI, Matplotlib and Seaborn.